Chapter 2: Second Quantization:

Exercise #1: (2.8) "Coordinate Basis for Transformation Law"

$$a^\dagger_{\tilde{\lambda}} = \sum_{\lambda} \langle \lambda | \tilde{\lambda} \rangle a^\dagger_\lambda \quad a_{\tilde{\lambda}} = \sum_{\lambda} \langle \tilde{\lambda} | \lambda \rangle a_\lambda$$

where $|\tilde{\lambda}\rangle = a^\dagger_\lambda |0\rangle$ and $|\tilde{\lambda}\rangle = a^\dagger_{\tilde{\lambda}} |0\rangle$

(pg 46) "Transformation formula"

$$a^\dagger_{\tilde{\lambda}} = \sum_{\lambda} \langle \lambda | \lambda \rangle a^\dagger_\lambda \quad a_\lambda = \int dx \langle \lambda | x \rangle a(x)$$

$$a_k = \int_0^L dx \langle k | x \rangle a(x)$$ ... where $\langle k | x \rangle = \langle x | k \rangle^{*} = \frac{e^{-ikx}}{\sqrt{L}}$

$$= \int_0^L dx \langle k | x \rangle \langle x | k \rangle a_k$$

$$= \int_0^L dx \frac{e^{-ikx}}{\sqrt{L}} \frac{e^{-ikx}}{\sqrt{L}}$$

$$= \frac{1}{L} \int_0^L dx a_k$$

$$= a_k$$

Exercise #2: (pg 21) "Hamiltonian of a Harmonic Oscillator"

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2$$

(pg 48) "Momentum Operator"

$$\hat{p} = -i\hbar\,\partial_x$$

$$\hat{H} = \frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}\hat{x}^2$$

$$= \int d^d r \langle r|\bar{r}\rangle\langle\bar{r}|a\rangle\left(\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}\hat{x}^2\right)\langle\bar{r}|r\rangle\langle r|\bar{r}\rangle$$

$$= \int d^d r \langle r|\bar{r}\rangle a_r^\dagger\left(\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}\hat{x}^2\right)\langle\bar{r}|r\rangle a_r$$

$$= \int d^d r\, a^\dagger(r)\left(\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}\hat{x}^2\right)a(r).$$

(2.14) "One-body Hamiltonian"

$$\hat{H} = \int d^d r\, a^\dagger(r)\left[\frac{\hat{p}^2}{2m} + V(r)\right]a(r)$$

Equation #3: (pg 55) "Wannier representation of the Hamiltonian"

$$\hat{H} = \sum_k \epsilon_k\, a^\dagger_{k\sigma} a_{k\sigma} = \frac{1}{N}\sum\sum e^{ik(R_i - R_{i'})}\epsilon_k a^\dagger_{i\sigma} a_{i'\sigma} = \sum_{ii'} a^\dagger_{i\sigma'} t_{ii'} a_{i\sigma}$$

$$\text{where } t_{ii'} = N^{-1}\sum_k e^{ik(R_i - R_{i'})}\epsilon_k$$

(2.24) "Block and Wannier States"

$$a^\dagger_{k\sigma} = \frac{1}{\sqrt{N}}\sum_i e^{ik\cdot R_i} a^\dagger_{i\sigma} \qquad a^\dagger_{i\sigma} = \frac{1}{\sqrt{N}}\sum e^{-ik_i} a^\dagger_{k\sigma}$$

<u>Square Lattice Geometry:</u>

$$\hat{H}_0 = \sum_{ii'} a^\dagger_{i\sigma} t_{ii'} a_{i'\sigma}$$

$$= \sum_{i=-1} \sum_{i'=-1} a_{i\sigma}^{\dagger} t_{ii'} a_{i'\sigma}$$

① Matrix:

$$\mathrm{diag}(\hat{H}_0) = \mathrm{diag}\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} t e^{ikx}$$

... when $t_{ii'} = \begin{cases} -t & i \neq |i'| \\ 0 & \text{otherwise} \end{cases}$

② Eigenvalue:

$$\hat{H}_0 - \lambda = \begin{pmatrix} -\lambda & -1 & 0 \\ -1 & -\lambda & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$= -\lambda^3 + 2\lambda$$

$$= 0$$

$$\lambda_1 = -\sqrt{2} \quad \lambda_2 = 0 \quad \lambda_3 = +\sqrt{2}$$

③ Eigenvectors:

$\lambda_1 = -\sqrt{2}$

$$H - \lambda_1 = \begin{pmatrix} \sqrt{2} & -1 & 0 \\ -1 & \sqrt{2} & -1 \\ 0 & -1 & \sqrt{2} \end{pmatrix} t e^{ikx}$$

$$(H-\lambda)X = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -\sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} -1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$\underline{\lambda_3 = -\sqrt{2}}$

$$\hat{H}_0 - \lambda_3 = \begin{pmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{pmatrix} t e^{ikx}$$

$$(H-\lambda_3)X = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$\underline{\lambda_2 = 0:}$

$$H - \lambda_2 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} t e^{ikx}$$

$$(H - \lambda_2)X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

④ Eigenvectors are linearly independent:

$$D = \text{eigenvalues}(\hat{H})$$

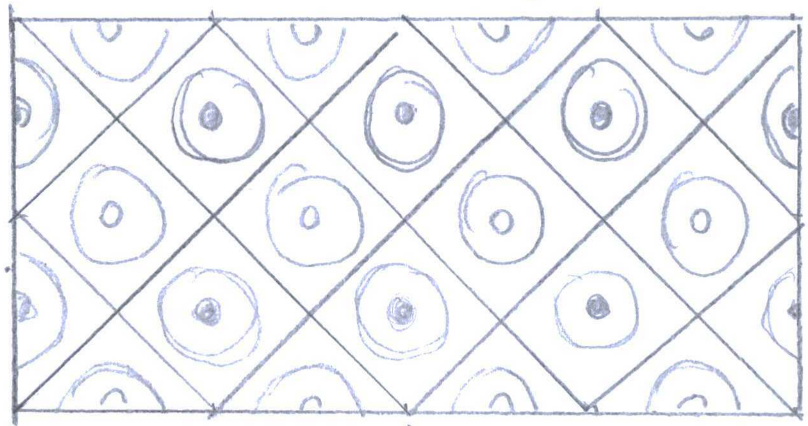
$$= \begin{pmatrix} -\sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} t^{ikx}$$

Function and Contours:

$$\hat{H} = t\left(e^{ik_x a} + e^{-ik_x a} + e^{ik_y a} + e^{-ik_y a}\right)$$

$$= -2t(\cos(k_x a) + \cos(k_y a))$$

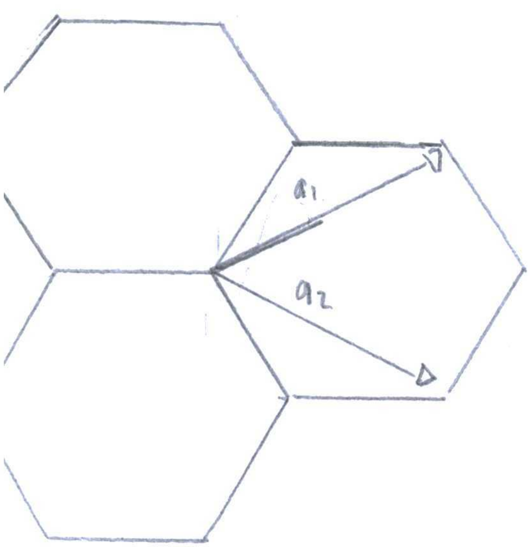
$= 0$, when $k_x = k_y$



"Brillioun Zone"

Half-filling geometry seperates occupancy and unoccupied bands.

Exercise #4:



"Reciprocal lattice vectors"

Cite: Gray, D., Mookerji, B.
"Crystal structure of Graphite, Graphene, and Silicon"

$$b_1 = \frac{2\pi a_1 \times a_3}{a_1 \cdot a_3 \times a_3} \qquad b_2 = \frac{2\pi a_3 \times a_3}{a_2 \cdot a_3 \times a_1}$$

$$= \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -1, 0\right) \qquad = \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, 1, 0\right)$$

$$G_{1/2} = b_1, b_2 = \frac{2\pi}{\sqrt{3}a}\left(1, \pm\sqrt{3}\right)$$

Fourier decomposition field operator:

(2.24) "operator transform"

$$a^{\dagger}_{k\sigma} = \frac{1}{\sqrt{N}} \sum_{k} e^{ikR_i} a^{\dagger}_{k\sigma}$$

$$a_k^{\dagger}(r) = \frac{1}{\sqrt{N}} \sum_k e^{ikR} a_{k\sigma}^{\dagger}$$

$$a_k(r) = \frac{1}{\sqrt{N}} \sum_k e^{\frac{-ia}{2\pi}(k_1 G_1 + k_2 G_2)} a_{k\sigma}$$

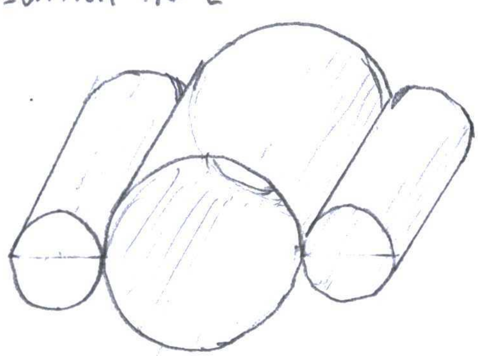
(2.25) "Fourier Representation"

$$\hat{H} = \sum [a_{1\sigma}^{\dagger} \; a_{2\sigma}^{\dagger}] \begin{bmatrix} 0 & -tf(k) \\ -tf^{*}(k) & 0 \end{bmatrix} \begin{bmatrix} a_{1\sigma} \\ a_{2\sigma} \end{bmatrix}$$

...where $f(k) = 1 + e^{-ik_1 a} + e^{i(-k_1+k_2)a}$

$$= \sum [a_a^{\dagger}(r) \; a_a^{\dagger}(r)] \begin{bmatrix} 0 & -tf(k) \\ -tf^{*}(k) & 0 \end{bmatrix} \begin{bmatrix} a_a(r) \\ a_a(r) \end{bmatrix}$$

Exercise removed from: Edition no. 2



"armchair graphene tube"

(2.26) "Dispersion relation"

$$\epsilon_k = \pm t\,|f(k)| = \pm t\sqrt{3 + 2\cos(k_1 a) + 2\cos((k_1 - k_2)a) + 2\cos(k_2 a)}$$

$$= \pm t\sqrt{3 + 4\cos\left(\frac{k_1+k_2}{2}a\right)\cos\left(\frac{k_1-k_2}{2}a\right) + 2\cos(k_{\parallel} a)}$$

$$= \pm t\sqrt{3 + 4\cos(0)\cos\left(\frac{k_{\parallel} a}{2}\right) + 2\cos(k_{\parallel} a)}$$

Trigonometric Identity - Sum to Product
$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$

... when $k_\perp = k_1 + k_2 = 0$

$$\mp \pm t\sqrt{3 + 4\cos(k_\parallel a/2) + 2\cos(k_\parallel a)}$$

(pg 58) in edition no. 2.

<u>Quantization rule:</u>

(2.22) "Wannier states"

$$|\psi_{rn}\rangle = \frac{1}{\sqrt{N}} \sum_R e^{ikR} |\psi_{Rn}\rangle$$

$$|\psi(r + Na\sqrt{3})\rangle = \frac{1}{\sqrt{N}} \sum_R e^{ik(r+Na\sqrt{3})} |\psi_{kn}\rangle$$

$$= \frac{1}{\sqrt{N}} \sum_R e^{i(k_1+k_2)(r+Na\sqrt{3})} |\psi_{kn}\rangle$$

$$= \frac{1}{\sqrt{N}} \sum e^{i(k_1+k_2)(r+L)} |\psi_{kn}\rangle$$

$$e^{i(k_1+k_2)(r+Na\sqrt{3})} = e^{i(k_1+k_2)(r+L)}$$

$$L = Na\sqrt{3}$$

and

$$(k_1+k_2) L = \frac{2\pi m}{L}$$

<u>Euler's</u> <u>Identity:</u>

$$e^{i\pi} + 1 = 0$$

$$e^{2\pi i} = 1$$

The book has a $\sqrt{3}$ coefficient not in the above answer. Also, the author in second edition supported zero-energy excitation. Unsure about zero-energy.

<u>Cite:</u> R. Saito, Dresselhaus (Wife & Husband) "Physical properties of Carbon Nanotubes" University of Electro-Communications, Tokyo. Massachusetts Institute of Technology.

Zig-zag tubes relate to gas and liquid sensors through resistance (ohm) during exposure.

Exercise #6: (from other sources) "Hamiltonian with Pauli matrices"

$$\hat{H} = \hbar v_F(\sigma_x k_x + \sigma_y k_y)$$

where V_k = fermi velocity

k = wave vector

σ_{xy} = Pauli matrices

$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

Exercise #5: (2.25) "Hamiltonian"

E_{TOTAL}
↓
$\sum_i E_i$
↓
$\sum_i \hat{H}_i$
↓
$\sum$ "transitions"

$$\hat{H} = \sum \begin{bmatrix} a^\dagger_{1\sigma} & a^\dagger_{2\sigma} \end{bmatrix} \begin{bmatrix} 0 & -t f(q) \\ -t f^*(q) & 0 \end{bmatrix} \begin{bmatrix} a_{1\sigma} \\ a_{2\sigma} \end{bmatrix}$$

$$= \underbrace{-a^\dagger_{20}\, t f^*(q)\, a_{20}}_{\hat{H}_1} - \underbrace{a^\dagger_{10}\, t f(k)\, a_{10}}_{\hat{H}_2}$$

Exercise #6: (from problem) "Pauli matrix identity"

$$\sigma_{\alpha\beta} \cdot \sigma_{\gamma\delta} = 2\delta_{\alpha\beta} \cdot \delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}$$

(from problem) "Spin operator for ½"

$$\hat{S}_i = \frac{1}{2} \sum_{\alpha,\beta} a^\dagger_{i\alpha} \sigma_{\alpha\beta} a_{i\beta}$$

$$\hat{S}_i \cdot \hat{S}_j = (a^\dagger_{i\alpha} \sigma_{\alpha\beta} a_{i\beta}/2)(a^\dagger_{j\gamma} \sigma_{\gamma\delta} a_{j\delta}/2)$$

$$= \frac{1}{4} a^\dagger_{i\alpha} a^\dagger_{j\gamma} (\sigma_{\alpha\beta} \cdot \sigma_{\gamma\delta}) a_{i\beta} \cdot a_{j\delta}$$

$$= \frac{1}{4} a^\dagger_{i\alpha} a^\dagger_{j\alpha} (2\delta_{\alpha\delta} \cdot \delta_{\beta\gamma} - \delta_{\alpha\beta}) a_{i\beta} \cdot a_{j\delta}$$

$$= a^\dagger_{i\alpha} a^\dagger_{j\gamma} a_{i\beta} a_{j\delta} / 2 - a^\dagger_{i\alpha} a^\dagger_{i\beta} a_{i\beta} a_{i\delta} / 4$$

$$= - a^\dagger_{i\alpha} a^\dagger_{j\beta} a_{i\beta} a_{j\alpha} / 2 - \hat{n}_i \cdot \hat{n}_j / 4$$

(pg 60). "Pauli matrix identity"

$$\sum_{i \neq j} U_{ijji} a^\dagger_{i\sigma} a^\dagger_{j\sigma'} a_{i\sigma'} a_{j\sigma} = -2 \sum J^F_{ij} (S_i \cdot S_j + \tfrac{1}{4} \hat{n}_i \hat{n}_j)$$

"Energy" $\propto$ "Ferromagnetic coupling" $\times$ ("spin" + "state")

Exercise #7: (2.30) "Cannonical Transformation to Hamiltonian"

$$\hat{H} \to \hat{H}' \equiv e^{-t\hat{O}} \cdot \hat{H} \cdot e^{t\hat{O}} = e^{-t[\hat{O},]} \hat{H} = \hat{H} - t[\hat{O}, \hat{H}] + \frac{t^2}{2!} [\hat{O}, [\hat{O}, \hat{H}]]$$

<u>Second equality:</u>

$$e^{-t[\hat{O},]} \hat{H} = (1 - t[\hat{O},] + \frac{t^2}{2!} [\hat{O}, [\hat{O},]]) \hat{H}$$

$$= \hat{H} - t[\hat{O}, \hat{H}] + \frac{t^2}{2!} [\hat{O}, [\hat{O}, \hat{H}]]$$

<u>Taylor expansion for an exponential</u>

$$e^X = 1 + X + \frac{X^2}{2!}$$

Exercise #8: (12.31) "Hamiltonian expanded"

Definition:
Ansatz - an assumption about a function facilitating a solution to an equation.

$$\hat{H}' = \hat{H}_v + \frac{t}{2}\left[\hat{H}_t, \hat{O}\right] + O(t^3)$$

(pg 621) "Ansatz"

$$t\hat{O} = \left[\hat{P}_s \hat{H}_t \hat{P}_d - \hat{P}_d \hat{H}_t \hat{P}_s\right]$$

(pg 62) "Projection Operators"

$\hat{P}_s$, $\hat{P}_d$

Singly occupied   doubly occupied

$$\hat{P}_s \hat{H}' \hat{P}_s = \hat{P}_s \left(H_v + \frac{t}{2}\hat{H}_t \hat{O} - \left(\frac{t}{2}\hat{O}\hat{H}_t\right)\right)\hat{P}_s$$

$$= \hat{P}_s \hat{H}_u \hat{P}_s + \hat{P}_s \frac{t}{2}\hat{H}_t \hat{O} \hat{P}_s - \frac{\hat{P}_s}{2}\left(\frac{\hat{P}_s \hat{H} \hat{P}_d - \hat{P}_d \hat{H}_t \hat{P}_s}{U}\right)\hat{H}_t \hat{P}_s$$

$$= \underbrace{\hat{P}_s \hat{H}_U \hat{P}_s}_{\hat{H}_U \hat{P}_s = 0} - \underbrace{\frac{P_s H_t \hat{O} P_s}{2}}_{\langle P_s | H_t \hat{O} | P_s \rangle = 0} - \frac{\hat{P}_s \hat{P}_s \hat{H}_t \hat{P}_d \hat{H}_t \hat{P}_s}{2U} + \underbrace{\frac{\hat{P}_s \hat{P}_d \hat{H}_t \hat{P}_s \hat{H}_t \hat{P}_s}{2U}}_{\hat{P}_s \circ \hat{P}_d = 0}$$

"P_s are eigenvalues to $\hat{H}_U$"   "Annhilation"   "orthogonal vectors"

$$= -\frac{\hat{P}_s \hat{H}_t \hat{P}_d \hat{H}_t \hat{P}_s}{U}$$

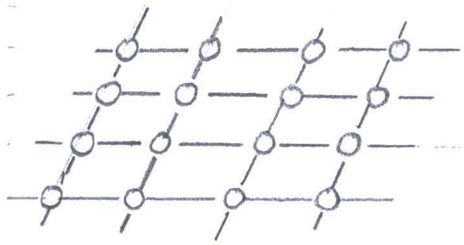
$$= 0$$

$$\ddot{H}' = -\frac{H_t P_d H_t}{U}$$

Exercise #9:

(Equation 2.29) "Hubbard Model"

$$\hat{H} = -t \sum_{\langle ij \rangle}' a^{\dagger}_{i\sigma} a_{j\sigma} + U \sum_{i}' \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$



where t = representation about state [±] or no state [0]

U = Hubbard interaction for Coulomb force

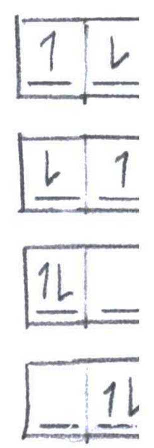
"Hubbard Model"

where o = $\Delta U \left\{ \begin{array}{l} \_\_ \; n=2 \\ \underline{1} \; n=1 \end{array} \right.$

(Pg 62) "Basis states"

$$\left.\begin{array}{l} |s_1\rangle = a^{\dagger}_{1\uparrow} a^{\dagger}_{2\downarrow} |\Omega\rangle \\ |s_2\rangle = a^{\dagger}_{2\uparrow} a^{\dagger}_{1\downarrow} |\Omega\rangle \end{array}\right\} \text{"singly occupied subspace"}$$

$$\left.\begin{array}{l} |d_1\rangle = a^{\dagger}_{1\uparrow} a^{\dagger}_{1\downarrow} |\Omega\rangle \\ |d_2\rangle = a^{\dagger}_{2\uparrow} a^{\dagger}_{2\downarrow} |\Omega\rangle \end{array}\right\} \text{"doubly occupied subspace"}$$



The Hubbard model describes lowest occupied atomic orbitals in ground state, with an index to each state. Hunds rule

Aufbau and Pauli exclusion principle apply.

$$\hat{H} = -t \begin{bmatrix} a^\dagger_{1\sigma} a_{1\sigma} & a^\dagger_{1\sigma} a_{2\sigma} & a^\dagger_{1\sigma} a_{3\sigma} & a^\dagger_{1\sigma} a_{4\sigma} \\ a^\dagger_{2\sigma} a_{1\sigma} & a^\dagger_{2\sigma} a_{2\sigma} & a^\dagger_{2\sigma} a_{3\sigma} & a^\dagger_{2\sigma} a_{4\sigma} \\ a^\dagger_{3\sigma} a_{1\sigma} & a^\dagger_{3\sigma} a_{2\sigma} & a^\dagger_{3\sigma} a_{3\sigma} & a^\dagger_{3\sigma} a_{4\sigma} \\ a^\dagger_{4\sigma} a_{1\sigma} & a^\dagger_{4\sigma} a_{2\sigma} & a^\dagger_{4\sigma} a_{3\sigma} & a^\dagger_{4\sigma} a_{4\sigma} \end{bmatrix} + U \begin{bmatrix} \hat{n}_{1\uparrow} \hat{n}_{1\downarrow} \\ \hat{n}_{2\uparrow} \hat{n}_{2\downarrow} \\ \hat{n}_{3\uparrow} \hat{n}_{3\downarrow} \\ \hat{n}_{4\uparrow} \hat{n}_{4\downarrow} \end{bmatrix}$$

$$= -t \begin{bmatrix} \langle d_1 | d_1 \rangle & \langle s_1 | s_2 \rangle & \langle s_1 | s_3 \rangle & \langle s_1 | s_4 \rangle \\ \langle s_2 | s_1 \rangle & \langle d_2 | d_2 \rangle & \langle s_2 | s_3 \rangle & \langle s_2 | s_4 \rangle \\ \langle s_3 | s_1 \rangle & \langle s_3 | s_2 \rangle & \langle d_3 | d_3 \rangle & \langle s_3 | s_4 \rangle \\ \langle s_4 | s_1 \rangle & \langle s_4 | s_2 \rangle & \langle s_4 | s_3 \rangle & \langle d_4 | d_4 \rangle \end{bmatrix} + U \begin{bmatrix} n_{1\uparrow} n_{1\downarrow} \\ n_{2\uparrow} n_{2\downarrow} \\ n_{3\uparrow} n_{3\downarrow} \\ n_{4\uparrow} n_{4\downarrow} \end{bmatrix}$$

$$= -t \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} + U \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

① <u>Matrix</u>

$$= \begin{bmatrix} U & -t & 0 & 0 \\ -t & U & -t & 0 \\ 0 & -t & U & -t \\ 0 & 0 & -t & U \end{bmatrix}$$

Note: 1's and 0's depend on the transition rules for fermions.

② Eigenvalues:

$$\hat{H}-\lambda = \begin{bmatrix} U-\lambda & -t & 0 & 0 \\ -t & U-\lambda & -t & 0 \\ 0 & -t & U-\lambda & -t \\ 0 & 0 & -t & U-\lambda \end{bmatrix}$$

$$= 0$$

$$-Ut^2(U-\lambda)+\lambda t^2(U-\lambda)-(t^2-(U-\lambda)^2)\cdot(U^2-2U\lambda+\lambda^2-t^2)$$

$$= 0$$

$$\lambda_1 = U+\frac{t}{2}(1-\sqrt{5}) \;, \quad \lambda_2 = U+\frac{t}{2}(1+\sqrt{5})$$

$$\lambda_3 = U-\frac{t}{2}(1+\sqrt{5}) \;, \quad \lambda_4 = U-\frac{t}{2}(1-\sqrt{5})$$

③ Eigenvectors:

$$\underline{\lambda_1 = U-\frac{t}{2}(1-\sqrt{5})}$$

$$\hat{H}-\lambda_1 = \begin{bmatrix} -\frac{t}{2}(1-\sqrt{5}) & -t & 0 & 0 \\ -t & -\frac{t}{2}(1-\sqrt{5}) & -t & 0 \\ 0 & -t & -\frac{t}{2}(1-\sqrt{5}) & -t \\ 0 & 0 & -t & -\frac{t}{2}(1-\sqrt{5}) \end{bmatrix}$$

$$(H-\lambda)X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1+\sqrt{5}}{2} \\ 0 & 0 & 1 & \frac{1-\sqrt{5}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} -t \\ \frac{t(1-\sqrt{5})}{2} \\ -\frac{t(1-\sqrt{5})}{2} \\ t \end{bmatrix}$$

$$\underline{\lambda_2 = U + \frac{t}{2}(1+\sqrt{5});}$$

$$\hat{H} - \lambda_2 = \begin{bmatrix} -\frac{t}{2}(1+\sqrt{5}) & -t & 0 & 0 \\ -t & -\frac{t}{2}(1+\sqrt{5}) & -t & 0 \\ 0 & -t & \frac{-t}{2}(1+\sqrt{5}) & -t \\ 0 & 0 & -t & \frac{-t}{2}(1+\sqrt{5}) \end{bmatrix}$$

$$(H-\lambda_2)X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & \frac{-1+\sqrt{5}}{2} \\ 0 & 0 & 1 & \frac{1+\sqrt{5}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} -t \\ \frac{t(1+\sqrt{5})}{2} \\ \frac{-t(1+\sqrt{5})}{2} \\ t \end{bmatrix}$$

$\underline{\lambda_3 = U - \frac{t}{2}(1+\sqrt{5})}$:

$$\hat{H} - \lambda_3 = \begin{bmatrix} \frac{t}{2}(1+\sqrt{5}) & -t & 0 & 0 \\ -t & \frac{t}{2}(1+\sqrt{5}) & -t & 0 \\ 0 & -t & \frac{t}{2}(1+\sqrt{5}) & -t \\ 0 & 0 & -t & \frac{t}{2}(1+\sqrt{5}) \end{bmatrix}$$

$$(H-\lambda_3)X = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{2}{1-\sqrt{5}} \\ 0 & 0 & 1 & \frac{2}{1-\sqrt{5}} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=\begin{bmatrix}t\\ \frac{t(1+\sqrt{5})}{2}\\ \frac{t(1+\sqrt{5})}{2}\\ t\end{bmatrix}$$

$\underline{\lambda_4 = U + \frac{t}{2}(1-\sqrt{5}):}$

$$\bar{H}-\lambda_4 = \begin{bmatrix}\frac{-t}{2}(1-\sqrt{5}) & -t & 0 & 0\\ -t & \frac{-t}{2}(1-\sqrt{5}) & -t & 0\\ 0 & -t & \frac{-t}{2}(1-\sqrt{5}) & -t\\ 0 & 0 & -t & \frac{-t}{2}(1-\sqrt{5})\end{bmatrix}$$

$$(H-\lambda_4)x = \begin{bmatrix}1 & 0 & 0 & -1\\ 0 & 1 & 0 & \frac{-1+\sqrt{5}}{2}\\ 0 & 0 & 1 & \frac{-1+\sqrt{5}}{2}\\ 0 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$$

$$=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} t \\ t\frac{(1-\sqrt{5})}{2} \\ t\frac{(1-\sqrt{5})}{2} \\ t \end{bmatrix}$$

④ Eigenvalues are linearly independent:

D = eigen values $(\hat{H})$

$$= \begin{bmatrix} U+\frac{t}{2}(1-\sqrt{5}) & 0 & 0 & 0 \\ 0 & U+\frac{t}{2}(1+\sqrt{5}) & 0 & 0 \\ 0 & 0 & U-\frac{t}{2}(1+\sqrt{5}) & 0 \\ 0 & 0 & 0 & U-\frac{t}{2}(1-\sqrt{5}) \end{bmatrix}$$

= "eigen states"

Strong coupling: $(U/t \gg 1)$:

$$\hat{H}_{gs} = \begin{bmatrix} U/t & -1 & 0 & 0 \\ -1 & U/t & -1 & 0 \\ 0 & -1 & U/t & -1 \\ 0 & 0 & -1 & U/t \end{bmatrix}$$

Exercise in edition no. 2 : a) (2.28) "Hubbard Model"

$$\hat{H} = -t\sum_{\langle ij\rangle\sigma} a^{\dagger}_{i\sigma} a_{j\sigma} + \sum_{i} \hat{n}_{i\uparrow}\hat{n}_{j\downarrow}$$

$$= -t\left[2\cdot a_{2\uparrow}a_{1\uparrow} \quad 2\cdot a_{1\downarrow}a_{2\downarrow} \quad 2\cdot a_{2\uparrow}a_{3\uparrow} + 2a_{3\downarrow}a_{2\downarrow} + 2\cdot a_{3\uparrow}a_{4\uparrow} + 2\cdot a_{4\downarrow}a_{3\downarrow}\right]$$

$$+ U[0]$$

... from "combinations":

$$\binom{\text{states}}{\text{electrons}} = \binom{4}{3} = \frac{4!}{3!\,(4-3)!}$$

... or pictures:

$\uparrow$	$\uparrow$	$\uparrow$	_
_	$\uparrow$	$\uparrow$	$\uparrow$
$\uparrow$	_	$\uparrow$	$\uparrow$
$\uparrow$	$\uparrow$	_	$\uparrow$
$\downarrow$	$\downarrow$	$\downarrow$	_
$\downarrow$	$\downarrow$	_	$\downarrow$
$\downarrow$	_	$\downarrow$	$\downarrow$
_	$\downarrow$	$\downarrow$	$\downarrow$

$$= \begin{bmatrix} 0 & -2t & 0 & 0 \\ -2t & 0 & -2t & 0 \\ 0 & -2t & 0 & -2t \\ 0 & 0 & -2t & 0 \end{bmatrix}$$

<u>Eigenspectrum</u>:

Magical Relationship

Erwin Schrodinger "Undulatory Theory" (1926)	$\hat{H}\Psi = E\Psi$
Cauchy "Indirect root" (1826)	$Ax = \lambda x$

Steps: ① Matrix:

$$\hat{H} = \begin{bmatrix} 0 & -2t & 0 & 0 \\ -2t & 0 & -2t & 0 \\ 0 & -2t & 0 & -2t \\ 0 & 0 & -2t & 0 \end{bmatrix}$$

② Eigenvalues:

$$\hat{H} - E = \begin{bmatrix} -E & -2t & 0 & 0 \\ -2t & -E & -2t & 0 \\ 0 & -2t & -E & -2t \\ 0 & 0 & -2t & -E \end{bmatrix} = 0$$

③ $E_1 = t(1-\sqrt{5})$ $\quad$ $E_2 = t(1+\sqrt{5})$

$E_3 = -t(1+\sqrt{5})$ $\quad$ $E_4 = -t(1+\sqrt{5})$

④ <u>Eigenvectors:</u>

$$(H-E_1)\psi = \begin{bmatrix} -t(1-\sqrt{5}) & -2t & 0 & 0 \\ -2t & -t(1-\sqrt{5}) & -2t & 0 \\ 0 & -2t & -t(1-\sqrt{5}) & -2t \\ 0 & 0 & -2t & -t(1-\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} 1 \\ t\,\frac{(1+\sqrt{5})}{2} \\ t\,\frac{(1+\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_2)\psi = \begin{bmatrix} -t(1+\sqrt{5}) & -2t & 0 & 0 \\ -2t & -t(1+\sqrt{5}) & -2t & 0 \\ 0 & -2t & -t(1+\sqrt{5}) & -2t \\ 0 & 0 & -2t & -t(1+\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} -1 \\ t\,\frac{(1+\sqrt{5})}{} \\ -t\,\frac{(1+\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_3)\psi = \begin{bmatrix} t(1+\sqrt{5}) & -2t & 0 & 0 \\ -2t & t(1+\sqrt{5}) & -2t & 0 \\ 0 & -2t & t(1+\sqrt{5}) & -2t \\ 0 & 0 & -2t & t(1+\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} -1 \\ -t\frac{(1-\sqrt{5})}{2} \\ -t\frac{(1-\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_4)\psi = \begin{bmatrix} t(1-\sqrt{5}) & -2t & 0 & 0 \\ -2t & t(1-\sqrt{5}) & -2t & 0 \\ 0 & -2t & t(1-\sqrt{5}) & -2t \\ 0 & 0 & -2t & t(1-\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} 1 \\ t\,\frac{(1-\sqrt{5})}{2} \\ t\,\frac{(1-\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

⑤ Eigenvectors are linearly independent, so

$$E_1 = t(1-\sqrt{5})$$

$$E_2 = t(1+\sqrt{5})$$

$$E_3 = -t(1+\sqrt{3})$$

$$E_4 = -t(1-\sqrt{5})$$

"minimum theoretical spectrum"

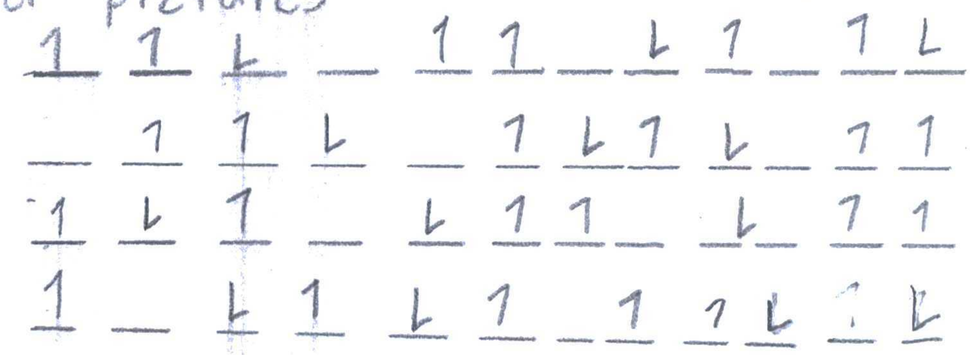
b) $S_{total} = 1/2$

Spin configurations:

.. by combinations:

$$\binom{\text{states}}{\text{electrons}} = \binom{4}{2} = \frac{4!}{2!(4-2)!}$$

... or pictures



Hubbard Model:

$$H = -t \sum_{\langle ij \rangle}^{4,4} a_{i\sigma}^{\dagger} a_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{j\downarrow}$$

$$= -t \sum_{\langle ij \rangle}^{4,4} (a_{i\uparrow}^{\dagger} a_{j\uparrow} - a_{i\downarrow}^{\dagger} a_{j\downarrow} - a_{i\uparrow}^{\dagger} a_{j\downarrow} - a_{i\uparrow}^{\dagger} a_{j\downarrow})$$

$$+ U \sum_{i} (\hat{n}_{i\uparrow} \hat{n}_{j\uparrow} + \hat{n}_{i\downarrow} \hat{n}_{j\downarrow})$$

$$= -t\Big(2(a_{1\uparrow}^{\dagger} a_{2\uparrow} + a_{1\downarrow}^{\dagger} a_{2\downarrow} + a_{1\uparrow}^{\dagger} a_{2\downarrow})$$

$$+ 2(a_{2\uparrow}^{\dagger} a_{1\uparrow} + a_{2\downarrow}^{\dagger} a_{1\downarrow} + a_{2\uparrow}^{\dagger} a_{1\downarrow})$$

$$+ 2(a_{2\uparrow}^{\dagger} a_{3\uparrow} + a_{2\downarrow}^{\dagger} a_{3\downarrow} + a_{2\uparrow}^{\dagger} a_{3\downarrow})$$

$$+ 2(a_{3\uparrow}^{\dagger} a_{2\uparrow} + a_{3\downarrow}^{\dagger} a_{2\downarrow} + a_{3\uparrow}^{\dagger} a_{2\downarrow})$$

$$+ 2(a_{3\uparrow}^{\dagger} a_{4\uparrow} + a_{3\downarrow}^{\dagger} a_{4\downarrow} + a_{3\uparrow}^{\dagger} a_{4\downarrow})$$

$$+ 2(a_{4\uparrow}^{\dagger} a_{3\uparrow} + a_{4\downarrow}^{\dagger} a_{3\downarrow} + a_{4\uparrow}^{\dagger} a_{3\downarrow})\Big)$$

$$+ U[0]$$

$$= \begin{bmatrix} 0 & -6t & 0 & 0 \\ -6t & 0 & -6t & 0 \\ 0 & -6t & 0 & -6t \\ 0 & 0 & -6t & 0 \end{bmatrix}$$

Eigenspectrum:

① Matrix:

$$\hat{H} = \begin{bmatrix} 0 & -6t & 0 & 0 \\ -6t & 0 & -6t & 0 \\ 0 & -6t & 0 & -6t \\ 0 & 0 & -6t & 0 \end{bmatrix}$$

(2) <u>Eigenvalues:</u>

$$\hat{H} - E = \begin{bmatrix} -E & -6t & 0 & 0 \\ -6t & -E & -6t & 0 \\ 0 & -6t & -E & -6t \\ 0 & 0 & -6t & -E \end{bmatrix}$$

$E_1 = 3t(1-\sqrt{5})$ $\quad E_2 = 3t(1+\sqrt{5})$

$E_3 = -3t(1+\sqrt{5})$ $\quad E_4 = -3t(1+\sqrt{5})$

(3) <u>Eigenvectors:</u>

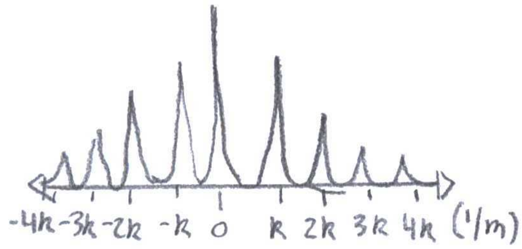
"... a time limit"

(4) <u>Eigenvectors are linearly independent:</u>

Eigenspectrum are the eigenvectors above.

Exercise #10: (2.35) "Hamiltonian recast"

$$V_{ee} = \frac{1}{2L} \sum_{kk'q}{}' V_{ee}(q)\, a^{\dagger}_{k-q}\, a^{\dagger}_{k'+q}\, a_{k'} a_k$$



"Second quantization"

$$= \frac{1}{2L} \sum_{qs} [g_4 \hat{\rho}_{sq} \hat{\rho}_{s-q} + g_2 \hat{\rho}_{sq} \hat{\rho}_{s-q}]$$

$$F.T. = \int_{-\infty}^{\infty} V_{ee}\, e^{-i\omega t}\, dt$$

$$= \frac{1}{2L} \int_{-\infty}^{\infty} [g_4 \hat{\rho}_{sq} \hat{\rho}_{s-q} + g_2 \hat{\rho}_{sq} \rho_{s-q}] e^{-i\omega t}\, dt$$

$$= \frac{1}{2L \cdot c} \int_{-\infty}^{\infty} [g_4 \hat{\rho}_{sq} \hat{\rho}_{s-q} + g_2 \hat{\rho}_{sq} \hat{\rho}_{s-q}] e^{ikx}\, dx$$

Summation $\sum_{kk'q}$:

$$(k, k', q) \simeq (\pm k_F, \pm k_F, 0)$$

$$V_{ee} = \frac{1}{2L} \sum_{kk'q} V_{ee}(0)\, a^\dagger_{kF} \cdot a^\dagger_{kF} \cdot a_{kF} \cdot a_{kF}$$

$$(k, k', q) \simeq (\pm k_F, \mp k_F, 2k_F)$$

$$V_{ee} = \frac{1}{2L} \sum_{kk'q} V_{ee}(2k_F) \cdot a^\dagger_{-kF}\, a^\dagger_{kF}\, a_{-kF}\, a_{kF}$$

Exercise #11: (2.36) "Commutator algebra"

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]_{\pm} \hat{C} \mp \hat{B} [\hat{A}, \hat{C}]_{\pm}$$

(pg 6.0) "Auxiliary identity to Commutation relation"

$$[\hat{\rho}_{sq}, \hat{\rho}_{s'q'}] = \delta_{ss'} \sum_k (a^\dagger_{sk+q} a_{sk-q'} - a^\dagger_{sk+q+q'} a_{sk})$$

$$= \delta_{ss'} \sum_k (a^\dagger_{sk+q} a_{sk+q} \cdot a_{-q} \cdot a_{-q'}$$
$$- a^\dagger_{sk} \cdot a^\dagger_{q} \cdot a^\dagger_{q'} \cdot a_{sk})$$

$$= \delta_{ss'} \sum_k (\hat{n}_{sk+q} \cdot a_{-q} \cdot a_{-q'}$$
$$- \hat{n}_{sk} \cdot a^\dagger_q a_{q})$$

$$= \delta_{ss'} \sum \langle \Omega | \hat{n}_{sk+q} \cdot a^\dagger_q a_{-q}$$
$$- \hat{n}_{sk} a^\dagger_q a_q)$$

$$\simeq \delta_{ss'} \delta_{q,-q'} \sum \langle \Omega | \hat{n}_{sk+q} - \hat{n}_{sk} | \Omega \rangle$$

... Where $\hat{n}_{sk} = a^\dagger_{sk} a_{sk}$

$\langle \Omega | a^\dagger_{sk} a_{sk'} | \Omega \rangle = \delta_{km}$

$\boxed{\text{\underline{Real observable}: energy difference, transition, or ground state.}}$

Exercise #12: <u>Expectation values:</u>

- Wavefunction: $|\Psi_{sq}\rangle \equiv \rho_{sq} |\Omega\rangle$

◦ Operator - $\hat{H}_0 = \hat{H}_0' + \text{const}$

◦ Transition - $\langle \Psi_{sq} | \hat{H}_0 | \Psi_{sq} \rangle = \langle \Psi_{sq} | \hat{H}_0' + \text{const} | \Psi_{sq} \rangle$

$$= \langle \Psi_{sq} | H_0' | \Psi_{sq} \rangle + \langle \Psi_{sq} | \text{const} | \Psi_{sq} \rangle$$

$$= \langle \Psi_{sq} | H_0' | \Psi_{sq} \rangle + \text{const} \langle \Omega | \hat{\rho}_{sq} \rho_{sq} | \Omega \rangle$$

$$= \langle \Psi_{sq} | H_0' | \Psi_{sq} \rangle$$

Exercise #13:

<u>Expectation values:</u>

◦ Operator - $\hat{S}' \equiv \sum_m \hat{s}_m'$

◦ Wavefunction - $|\alpha\rangle \equiv \exp(i\alpha \cdot \hat{s}) |\Omega\rangle$

◦ Transition - $\langle \alpha | \hat{S}' | \alpha \rangle = \langle \Omega | e^{-i\alpha \cdot \hat{s}} \cdot \hat{s} \cdot e^{i\alpha \hat{s}} | \Omega \rangle$

$$= \sum_m S_m' \langle \Omega | | \Omega \rangle$$

$$= \sum_m S_m'$$

<u>Energy Expectation:</u>

$$\langle \Omega | | \Omega \rangle = \int_{-\infty}^{\infty} S_m \otimes_m \hat{H} \otimes_m S_m \, d\tau$$

$$= \frac{1}{N} \int_{-\infty}^{\infty} S_m^2 \, d\tau$$

$$= \frac{E}{N} \quad \text{"average energy"}$$

(pg 74)
"Ground state"
$|\Omega\rangle = \otimes_m |S_m\rangle$

State $|(\pi/2, 0, 0)\rangle$:

If $\alpha = (\pi/2, 0, 0)$, then

$$\langle\alpha|\alpha\rangle = \langle\Omega| e^{-i\alpha s} \cdot e^{i\alpha s} |\Omega\rangle$$

$$= \langle S_m| \otimes e^{-i\pi/2 s} \cdot e^{i\pi/2 \cdot s} \otimes |S\rangle$$

Quantum Notation

- Wave function at a value

$$\langle 0.25|\psi\rangle = \psi(0.25)$$

- Outer product

$$|\phi\rangle\langle\psi| = \frac{|\phi_1\rangle + |\phi_2\rangle}{\sqrt{2}} \frac{|\psi_1\rangle + |\psi_2\rangle}{\sqrt{2}}$$

- Schrodinger equation

$$\hat{H}\psi = E\psi$$

- Expectation

$$\langle\psi|\hat{H}|\psi\rangle = \langle H\rangle$$

- Superposition

$$\langle\phi| = \frac{\langle\phi_1| + \langle\phi_2|}{2}$$

- Inner product

$\langle\psi|\phi\rangle$ = Dot product.

Exercise #14 (2.46) "Spin raising and lowering operators"

$$[\hat{S}_m^z, \hat{S}_n^{\pm}] = \pm \delta_{mn} \cdot S_m^{\pm} \quad [S_m^+, S_n^-] = 2\delta_{mn} S_m^{+}$$

(pg 75) "Holstein-Primakoff Transformation"

$$\hat{S}_m^- = a_m^\dagger (2S - a_m^\dagger a_m)^{1/2}$$

$$\hat{S}_m^+ = (2S - a_m^\dagger a_m)^{1/2} a_m$$

$$\hat{S}_m^z = S - a_m^\dagger a_m$$

<u>Identities from Quantum Mechanics:</u>

① Orbital Angular Momentum:

$$[S_x, S_y] = i\hbar S_z \,, \quad [S_y, S_z] = -i\hbar S_x \,, \quad [S_z, S_x] = i\hbar S_y$$

② Spin Angular Momentum:

$$S^2 = S_x^2 + S_y^2 + S_z^2$$

$$S^2 - S_z^2 = S_x^2 + S_y^2$$

$$= (S_x + iS_y)(S_x - iS_y)$$

$$= S_+ \circ S_-$$

$$[S^2, \hat{S}_\pm] = [S^2, S_x \pm iS_y]$$

$$= [S^2, S_x] \pm [S^2, iS_y]$$

$$= 0$$

$$[S_z, S_\pm] = S_z \circ S_\pm - S_\pm \cdot S_z$$

$$= [S_z, S_x \pm i S_y]$$

$$= [S_z, S_x] \pm [S_z, S_y]$$

$$= i\hbar S_y \pm i(-i\hbar S_x)$$

$$= \hbar(S_x + i S_y)$$

$$= \hbar S_+$$

$$[S_m^z, S_n^-] = [S_m^z, S_n^1 \pm i S_n^2]$$

$$= [S_m^z, S_n^1] \pm i [S_m^z, S_n^2]$$

$$= i\hbar S_m^2 \pm i(-i\hbar S_m^1)$$

$$= \hbar(S_m^1 \pm S_m^2)$$

$$= \pm\hbar S_m^\pm$$

$$= \pm\delta_{mn} S_m^\pm$$

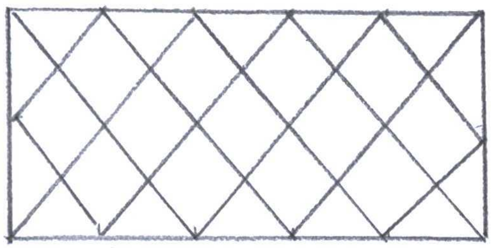
← "Book removed $\hbar$ as with Chapter 1 or on page 74"

$$[S_m^+, S_n^-] = [S_m^1 + i S_m^2, S_n^1 - i S_n^2]$$

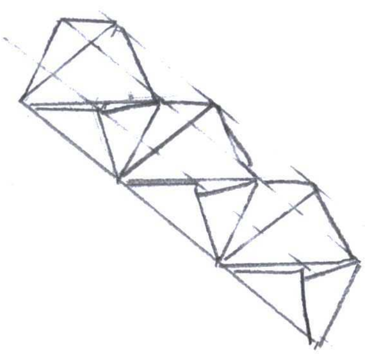
$$= \delta - a_m^\dagger a_n + \delta - a_m a_n^\dagger$$

$$= 2\delta_{mn} S_m^-$$

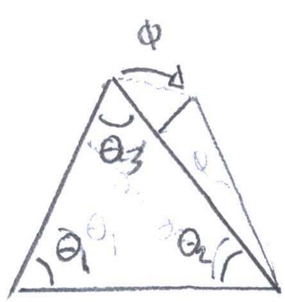
Exercise #15.



"Triangular lattice"



"Equilateral triangular pyramids lattice"



"equilateral triangle rotated"

(pg 76) "Heisenberg Antiferromagnet"

$$\hat{H} = J \sum_{\langle mn \rangle} \hat{S}_m \circ \hat{S}_n$$

$$= JS^2 \sum_{\langle mn \rangle} S_i S_m$$

$$= JS^2 (\cos\theta_1 \cos\phi_1 + \cos\theta_2 \cos\phi_2 + \cos\theta_3 \cos\phi_3)$$

.. when $\theta_1 = \frac{2\pi}{3}$ $\theta_2 = \frac{2\pi}{3}$ $\theta_3 = \frac{2\pi}{3}$

$\Phi_1 = 0$ $\Phi_2 = \frac{2\pi}{3}$ $\Phi_3 = \frac{\pi}{2}$

$$= -\frac{3}{2} JS^2$$

Final Energy:

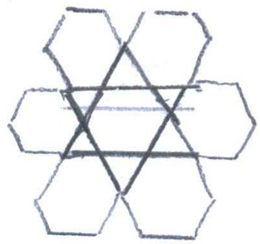
Schrodinger equation:

$$\hat{H}\Psi = -\frac{3}{2} JS^2 \Psi$$

$$= E\Psi$$

$$E = -\frac{3}{2} JS^2$$

Fun fact:
Book describes Ferromagnetic as negative energy and antiferromagnetic as positive.



"kagomé lattice"

Kagomé antiferromagnet:

Continuous spin degeneracy:

$$\phi_1 = \phi_2 + \phi$$

$$\phi_2 = \phi_3 + \phi$$

$$\phi_3 = \phi + \phi \quad \text{: where } \phi = 2\pi/3$$

$$\hat{H} = J \sum \hat{S}_m \cdot \hat{S}_n$$

$$= J S^2 (\cos\theta_1 \cos\phi_1 + \cos\theta_2 \cos\phi_2 + \cos\theta_3 \cos\phi_3)$$

$$= -\frac{3}{2} J S^2$$

Problem 2.4.1: Stone-Von Neumann Theorem:

a) Operator - $n_\lambda = a_\lambda^\dagger a_\lambda$

Wave function $|n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle$

Eigenvalues $\hat{n}|n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle = n|n_{\lambda_1}, n_{\lambda_2} \ldots\rangle$

Unique basis when a derivative provides no change to eigenvalues:

$$|n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle = P([a_\mu, a_\mu^\dagger])|n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle$$

$$= P[\hat{n}]\,|n_{\lambda_1}, n_{\lambda_2} \ldots\rangle$$

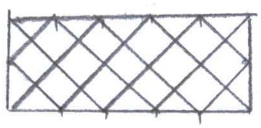
$$= n_\lambda\,|n_{\lambda_1}, n_{\lambda_2} \ldots\rangle$$

b) $\hat{n}_\lambda a_\lambda |n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle = (n-1)\,a\,|n_{\lambda_1}, n_{\lambda_2}, \ldots\rangle$

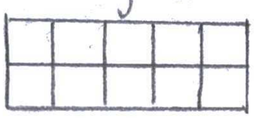
where $n-1 \geq 0$ and $n \geq 1$

Problem 2.4.2: Semi-classical spin waves:

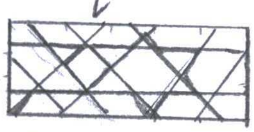
Lattices:



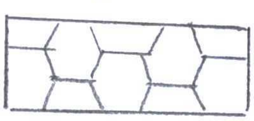
"Triangular"



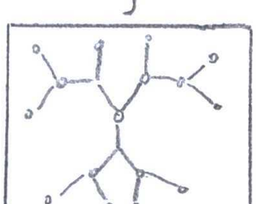
"Square"



"Kagomé"



"Honeycomb"



"Bethe"

a) (from Problem) "spin commutator relation"

$$[\hat{S}_i^\alpha, \hat{S}_j^\beta] = i\delta_{ij}\,\epsilon^{\alpha\beta\gamma}\,\hat{S}_i^\gamma$$

(from Problem) "Operator identity"

$$i\dot{\hat{S}}_i = [\hat{S}_i, \hat{H}]$$

(2.44) "Heisenberg ferromagnet"

$$\hat{H} = -J\sum_{\langle mn\rangle} \hat{S}_m \cdot \hat{S}_n$$

$$= -J\sum_{\langle mn\rangle} \hat{S}_m \cdot \hat{S}_{m+1} \quad \ldots \text{for one-dimensional ferromagnet}$$

$$[S_i, \hat{H}] = [\hat{S}_i, -J\sum \hat{S}_m \cdot \hat{S}_{m+1}]$$

$$= i\dot{\hat{S}}_i$$

Commutator relations:

$$[\hat{S}_i^\alpha, S_i^\beta \cdot S_{i+1}^\beta] = [\hat{S}_i^\alpha, \hat{S}_i^\beta]\,\hat{S}_{i+1}^\beta + \hat{S}_i\,[\hat{S}_i^\alpha, \hat{S}_{i+1}]$$

$$= i\epsilon^{\alpha\beta\gamma} \circ S_i^{\gamma} \cdot S_{i+1}^{\beta}$$

$$[S_i^{\alpha}, S_{i-1}^{\beta} \circ S_i^{\beta}] = [\hat{S}_i^{\alpha}, \hat{S}_{i-1}^{\beta}]\,\hat{S}_i^{\beta} + \hat{S}_{i-1}^{\beta}[\hat{S}_i^{\alpha}, \hat{S}_i^{\beta}]$$

$$= i\epsilon^{\alpha\beta\gamma} \cdot (\hat{S}_{i-1} \circ \hat{S}_i)_{\alpha}$$

$$[\hat{S}_i, \hat{H}_{TOT}] = [\hat{S}_i, \hat{H}_{i-1} + \hat{H}_{i+1}]$$

$$= [\hat{S}_i, -J\sum \hat{S}_{i-1} \circ \hat{S}_i - J\sum \hat{S}_i \hat{S}_{i+1}]$$

$$= [\hat{S}_i, -J\sum S_{i-1} \circ S_i] + [\hat{S}_i, -J\sum S_i S_{i+1}]$$

$$= -iJ[(S_{i-1} \cdot S_i) + (S_i \circ S_{i+1})]$$

$$= -i\hat{S}_i$$

$$\hat{S}_i = J \circ S_i \times (S_{i+1} + S_{i-1})$$

b) <u>Equation of Motion:</u>

$$\hat{S}_i = J \circ S_i \times (S_{i+1} + S_{i-1})$$

$$= J \circ S_i \times (S_i + \partial_x S_i + \tfrac{1}{2}\partial^2 S_i + S_i - \partial_x S_i + \tfrac{1}{2}\partial^2 S$$

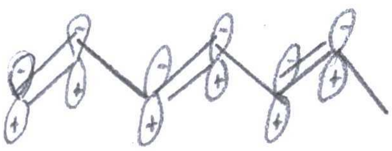
$$= J \circ \underbrace{S_i \times (2S_i}_{[S_i, S_i] = 0} + \partial^2 S_i)$$

$$= J \circ S_i \times \partial^2 S_i$$

.. edition no. 2 versus no. 3 mayhem!

Problem 2.4.3: Su-Shrieffer-Heeger model of a conducting polymer chain:

"Polyacetylene"



"sp^2-hybridized polymer chain"

"peieris distorted chain"

a) (from problem) "Effective Hamiltonian"

$$\hat{H} = -t\sum_{n=1,\sigma}^{N}(1+u_n)\left[c_{n\sigma}^{\dagger} c_{n+1\sigma} + h.c\right] + \sum_{n=1}^{N}\frac{K_s}{2}(u_{n+1}-u_n)^2$$

(from problem) "Lattice distortion"

$$u_n = (-1)^n \alpha$$

$$\hat{H}_{TOT} = \hat{H}_{n=even} + \hat{H}_{odd}$$

$$= -t\sum_{m=1,\sigma}^{N/2}\Big\{(1+\alpha)\left[a_{m\sigma}^{\dagger} b_{m\sigma} + h.c\right]$$

$$+ (1-\alpha)\left[b_{m\sigma}^{\dagger} a_{m+1\sigma} + h.c\right]\Big\}$$

$$+ 2NK_s\alpha^2$$

(2.24) "Operator Transformation" [Block and Wannier states]

$$a_{k\sigma}^{\dagger} = \frac{1}{\sqrt{N}}\sum_i e^{ikR_i} a_{i\sigma}^{\dagger}$$

$$a_{i\sigma}^{\dagger} = \frac{1}{\sqrt{N}}\sum_k e^{-ikR_i} \cdot a_{k\sigma}^{\dagger}$$

... where $a_i = \sqrt[2]{z_i}$

$$\hat{H} = -t\sum_{m=1,\sigma}^{N/2}\Big\{(1+\alpha)\Big[\sum_k^{\infty}\frac{2}{N}e^{2ikm} + h.c\Big] + (1-\alpha)\Big[\sum_k^{\infty}\frac{2}{N}e^{2ikm} + h.c.\Big]\Big\}$$

$$+ 2Nk\alpha^2$$

$$= -t\sum_k (a_{k\sigma}^{\dagger} \;\; b_{k\sigma}^{\dagger})\begin{pmatrix} 0 & (1+\alpha)+(1-\alpha)e^{2ikm} \\ (1+\alpha)+(1-\alpha)e^{-2ikm} & 0 \end{pmatrix}\begin{pmatrix} a_{k\sigma} \\ b_{k\sigma} \end{pmatrix}$$

Eigenspectrum:

① Matrix:

$$\hat{H} = \begin{pmatrix} 0 & (1+\alpha)+(1-\alpha)e^{2ikm} \\ (1+\alpha)+(1-\alpha)e^{-2ikm} & 0 \end{pmatrix}$$

② Eigenvalues:

$$\hat{H}-E = \begin{vmatrix} 2Nk\alpha^2 - E & 2Nk\alpha^2 + (1+\alpha)+(1-\alpha)e^{2ikm} \\ 2Nk\alpha^2 + (1+\alpha)+(1-\alpha)e^{-2ikm} & 2Nk\alpha^2 - E \end{vmatrix}$$

$$= 0$$

$$E_1 = +2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk\alpha^2$$

$$E_2 = -2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk\alpha^2$$

③ Eigenvectors:

... "on a time limit"

④ Eigenvectors are linearly independent.

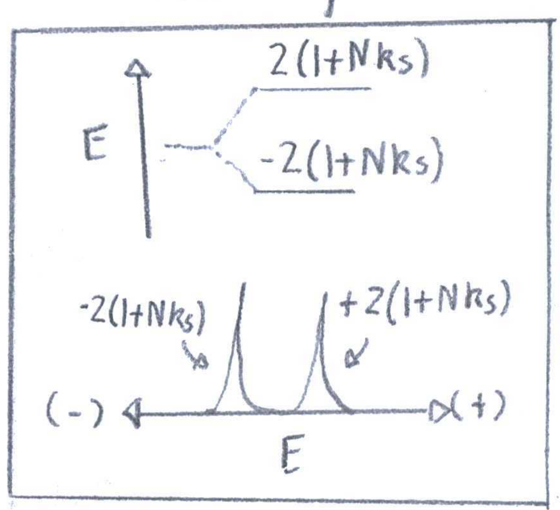
$$E_1 = 2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk_s\alpha^2$$

$$E_2 = -2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk_s\alpha^2$$

Distortion near Zero:

$$\lim_{\alpha \to 0} E = \lim_{\alpha \to 0}\left[\pm 2\sqrt{1+(\alpha^2-1)\sin^2 k}\right] - 2Nk_s\alpha^2$$

$$= \pm 2(1+Nk_s)$$ "A gap in the spectrum"



b)

(from Problem) "Formula"

$$\int_{-\pi/2}^{\pi/2} dk\left(1-(1-\alpha^2)\sin^2 k\right)^{1/2} \simeq 2 + (a_1 - b_1 \ln\alpha^2)\alpha^2 + O(\alpha^2 \ln^2 \alpha^2)$$

$$\int_{-\pi/2}^{\pi/2} E\,dk = \int_{-\pi/2}^{\pi/2} \pm 2t\sqrt{1+(\alpha^2-1)\sin^2 k} - 2NK_s\alpha^2 \, dk$$

$$= \pm 2t\left[2+(a_1-b_1\ln\alpha^2)\alpha^2\right] - 2NK_s\alpha^2]$$

<u>Maximum distortion:</u>

$$\frac{dE}{d\alpha} = \frac{d}{d\alpha}\left[\pm 2t\left[2+(a_1-b_1\ln\alpha^2)\right] - 2NK_s\alpha^2\right]$$

$$= \pm 2t\left[2a_1\cdot\alpha - b_1\left[\frac{2}{\alpha}\alpha^2 + 2\alpha^2\ln\alpha^2\right] - 4NK_s\alpha\right]$$

$$= 0$$

$$\alpha = e^{\frac{1}{2}(a_1/b_1 - 1 - NK_s/tb_1)}$$

Note: Peierls describes two spectra by two structures in a molecule.

c) The Peierls instability has two degenerate configurations; conjugated and unconjugated. An odd site number alters the structure and spectrum.

Problem 2.4.4: Schwinger boson representation:

(from problem) "Schwinger Boson representation"

$$\hat{S}^{+} = a^{\dagger} b \qquad \hat{S}^{-} = (\hat{S}^{+})^{\dagger} \qquad \hat{S}^{z} = \frac{1}{2}(a^{\dagger}a - b^{\dagger}b)$$

Commutation relation:

$$[\hat{S}^{+}, \hat{S}^{-}] = \hat{S}^{+} \cdot \hat{S}^{-} - \hat{S}^{-}\hat{S}^{+}$$

$$= a^{\dagger} b \cdot (\hat{S}^{+})^{\dagger} - (\hat{S}^{+}) a^{\dagger} b$$

$$= a^{\dagger} b - b^{\dagger} b$$

$$= 2 \cdot \hat{S}^{z}$$

b) If $\hat{S}^2 = (\hat{S}^{z})^2 + \frac{1}{2}(\hat{S}^{+}\hat{S}^{-} + b^{\dagger}a a^{\dagger} b)$

$$= \frac{1}{4}(a^{\dagger}a - b^{\dagger}b) + \frac{1}{2}(\hat{n}_a \cdot \hat{n}_b + \hat{n}_b n_a)$$

$$= \frac{1}{2}(\hat{n}_a + \hat{n}_b)^2$$

Total Spin:

$$\hat{S}^2 |S, m\rangle = S^2 |S, m\rangle$$

(from problem)

$$n_a + n_b | n_a, n_b\rangle = 2S$$

$$\hat{S}^2 |S, m\rangle = \frac{1}{4}(\hat{n}_a + \hat{n}_b)^2 |S, m\rangle$$

Spin operator:

$$\hat{S}^z|s,m\rangle = \frac{1}{2}(\hat{n}_a - n_b)|n_a = s, n_b = s-m\rangle$$
$$= m|s,m\rangle$$

Problem 2.4.5: Jordan-Wigner Transformation:

(from problem) "Spin lowering and raising operators"

$$\hat{S}^+ = f^\dagger \; ; \; \hat{S}^- = f \; ; \; \hat{S}^z = f^\dagger f - 1/2$$

a) Commutator relationship:

$$[\hat{S}^+, \hat{S}^-] = \hat{S}^+\hat{S}^- - \hat{S}^-\hat{S}^+$$
$$= f^\dagger f - f f^\dagger$$
$$= 2S^z$$

Real-world application: Spin operators apply to fermions in Electron Paramagnetic Resonance and bosons in Nuclear magnetic resonance.

Nonlinear Operator Transformation:

$$\hat{S}_\ell = f_\ell^\dagger \exp(i\pi \sum_{j<\ell} \hat{n}_j)$$
$$\hat{S}_\ell^- = \exp(-i\pi \sum_{j<\ell})f_\ell$$
$$\hat{S}_\ell^z = f_\ell^\dagger f_\ell - 1/2$$

$$[\hat{S}_\ell^+, \hat{S}_\ell^-] = \hat{S}_\ell^+\hat{S}_\ell^- - \hat{S}_\ell^- S_\ell^+$$
$$= f^\dagger \exp(+i\pi \sum_{j<\ell})\exp(-i\pi \sum_{j<\ell}) f_\ell - \exp(-i\pi \sum_{j<\ell})$$
$$\circ f_\ell \circ f_\ell^\dagger \exp(+i\pi \sum_{j<\ell})$$

$$= f_\ell^\dagger f_\ell - f_\ell f^\dagger$$
$$= 2f^\dagger f - 1$$
$$= 2S^z$$

b) $$\hat{S}_m^\dagger \hat{S}_{m+1}^- = f_m^\dagger \exp(i\pi \textstyle\sum_{j<m} n_j) \exp(-i\pi \sum_{\ell<m+1} n_\ell) f_{m+1}$$
$$= f_m^\dagger e^{-i\pi n_m} \circ f_{m+1}$$
$$= f_m^\dagger \circ f_{m+1}$$ ∘∘∘ when $n=2$ particles

c) (from problem) "Heisenberg Spin Chain"

$$\hat{H} = -\sum_n \left(J_z \hat{S}_n^z \hat{S}_{n+1}^z + \frac{J_\perp}{2}\left(\hat{S}_n^+ S_{n+1}^- + \hat{S}_n^- \hat{S}_{n+1}^+\right)\right)$$
$$\doteq -\sum_n \left(J_n \left(f_n^\dagger f_n - \tfrac{1}{2}\right)\left(f_{n+1}^\dagger f_{n+1} - \tfrac{1}{2}\right) + \frac{J_\perp}{2}\left(f_n^\dagger \circ f_{n+1} + f_{n+1}^\dagger f_n\right)\right)$$

$$\doteq -\sum_n \left(\frac{J_\perp}{2}\left(f_n^\dagger \circ f_{n+1} + h.c.\right) + J_z\left(\frac{1}{4} - f_n^\dagger \circ f_n + f_n^\dagger f_n f_{n+1}^\dagger f_{n+1}\right)\right)$$

d) If $J_z = 0$, then

$$\hat{H} = -\sum_n \left(\frac{J_\perp}{2}\left(f_n^\dagger f_{n+1} + h.c.\right)\right)$$
$$= -\sum \frac{J_\perp}{2} \begin{pmatrix} f_n^\dagger & f_{n+1} \end{pmatrix} \begin{pmatrix} 0 & e^{-2\pi i a} \\ e^{-2\pi i a} & 0 \end{pmatrix} \begin{pmatrix} f_n \\ f_{n+1} \end{pmatrix}$$

Eigenspectrum:

① **Matrix:**

$$\hat{H} = \begin{pmatrix} 0 & e^{-2\pi i a} \\ e^{-2\pi i a} & 0 \end{pmatrix}$$

② **Eigenvalues:**

$$(\hat{H} - E) = \begin{pmatrix} -E & e^{-2\pi i a} \\ e^{-2\pi i a} & -E \end{pmatrix}$$

$$= 0$$

$$E_1 = e^{-2\pi i a} \qquad E_2 = -e^{-2\pi i a}$$

③ **Eigenvectors:**

$$(\hat{H} - E_1) = \begin{pmatrix} -e^{-2\pi i a} & e^{-2\pi i a} \\ e^{-2\pi i a} & -e^{-2\pi i a} \end{pmatrix}$$

$$(\hat{H} - E_1)\psi = \begin{pmatrix} -e^{-2\pi i a} & e^{-2\pi i a} \\ e^{-2\pi i a} & -e^{-2\pi i a} \end{pmatrix} \begin{pmatrix} |\psi\rangle \\ |\psi\rangle \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$|\psi\rangle = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$(H - E_2)\psi = \begin{pmatrix} e^{-2\pi i a} & e^{-2\pi i a} \\ e^{-2\pi i a} & e^{-2\pi i a} \end{pmatrix} |\psi\rangle$$

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

$$E = -J\cos(Ka)$$

$$= 0$$

$$|\psi = |{}^{1}_{1}\rangle$$

④ Eigenvectors are linearly independent

$$Re(E_1) = \cos(2\pi k_1) = \cos(Ka)$$

Problem 2.4.6: Spin-charge separation in one-dimension:

Hydrodynamic photographs were unreal for spin-charge separation.

(2.34) "Operator"

$$\hat{\rho}_{sq} = \sum_k a^\dagger_{sk+q} a_{sk}$$

(from problem) "Operator"

$$\hat{\rho}_{sq\alpha} = \sum_k a^\dagger_{s(k+q)\alpha} \cdot a_{sk\alpha} \quad \ldots \text{where } \alpha = \uparrow, \downarrow$$

(2.39) "bosonic commutation relation"

$$b_q = n_q \hat{\rho}_{L-q} \qquad b_q^\dagger = n_q \hat{\rho}_{L-q}$$

$$b_{-q} = n_q \hat{\rho}_{R-q} \qquad b_{-q}^\dagger = n_q \hat{\rho}_{Rq}$$

(from problem) "spin index"

$$b_q \longrightarrow b_{q\alpha}$$

(from problem) "Momentum spin interaction"

$$|k_F + q + q_1, \uparrow; k_F + q - q_1, \downarrow\rangle$$

$$\longrightarrow |-k_F + q + q_2, \uparrow; k_F + q - q_2, \downarrow\rangle$$

(from problem) "Effective Bosonic Hamiltonian"

$$\hat{H} = \sum_{q>0,s,q} v_F \cdot q \cdot b^{\dagger}_{sq\alpha} \cdot b_{sq} + \sum_{q>0,s,\alpha\alpha'} |q| \left[\frac{g_2}{2\pi} \left(b^{\dagger}_{sq\alpha} b^{\dagger}_{sq\alpha'} + h.c \right) + \frac{g_4}{2\pi} b^{\dagger}_{sq\alpha} b_{sq\alpha'} \right]$$

(from problem) "Operators' Charge"

$$b_{sqe} = \frac{1}{\sqrt{2}} \left(b_{sq\uparrow} + b_{sq\downarrow} \right)$$

"Operator spin"

$$b_{sq\sigma} = \frac{1}{\sqrt{2}} \left(b_{sq\uparrow} - b_{sq\downarrow} \right)$$

$$\hat{H} = \sum_{q>0,s} v_F \cdot q \left(b^{\dagger}_{sqe} b_{sqe} + b^{\dagger}_{sq\sigma} b_{s\sigma} \right) + |q| \left[\frac{g_2}{\pi} \left(b_{sqe} b_{sqe} + \frac{g_4}{\pi} b^{\dagger}_{sqe} b_{sqe} + h.c \right] \right.$$

(pg 71) "Bogoliubov Transformation"

$$T = \begin{pmatrix} \cosh \theta_k & \sinh \theta_k \\ \sinh \theta_k & \cosh \theta_k \end{pmatrix}$$

where $\tanh(2\theta) = -g_2/(2\pi v_F + g_4)$

"The book confused me on page 71. Wikipedia provided a formula after $\hat{H}$-rearrangement"

$$\hat{H} = \sum_{q>0,s} \Big[\underbrace{\left(v_F |q| + \frac{g_4}{\pi}\right)}_{\text{"u"}} b_{sqe} + \underbrace{|q| \frac{g_2}{\pi}}_{\text{"v"}} b^{\dagger}_{sqe} \Big] b^{\dagger}_{sqe}$$

$$+ v_F q\, b^{\dagger}_{sq\sigma} b_{sq\sigma}$$

Wikipedia Bogoliubov transformation:

① Transformation matrix

$$U = \begin{pmatrix} u & v \\ v^* & u^* \end{pmatrix}$$

② Inverse matrix:

$$U^{-1} = \begin{pmatrix} u & v^* \\ v & u^* \end{pmatrix}$$

③ Gamma:

$$\Gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

④ Diagnolization:

$$U(\Gamma H)U^{-1} = \Gamma D$$

$$\hat{H} = \sum_{q>0,s} \Big[|q| \left(b_{sq} \;\; b^{\dagger}_{sqe} \right) \begin{pmatrix} \left(v_F + \frac{g_4}{\pi}\right)^2 - \left(\frac{g_2}{\pi}\right) & 0 \\ 0 & 0 \end{pmatrix} b^{\dagger}_{sqe}$$

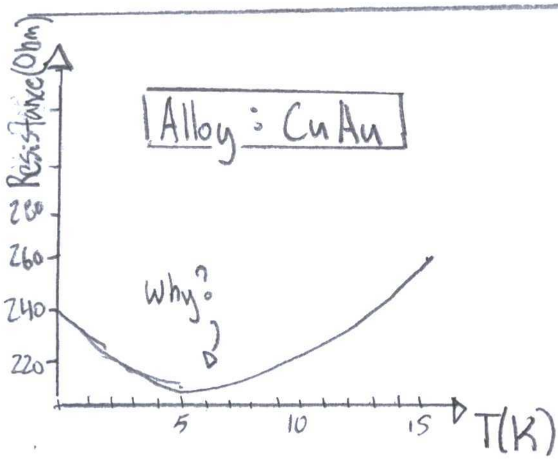
$$+ |q| v_F\, b^{\dagger}_{sq\sigma} b_{sq}$$

$$=\sum_{q>0,s} |q|\left(\left(v_F+\frac{g_4}{\pi}\right)^2+\left(\frac{g_2}{\pi}\right)^2\right) b^\dagger_{sql}\, b_{sql} + |q|\, v_F\, b^\dagger_{sq\sigma}\, b_{sq\sigma}$$

(Equation 2.7)

$$b^\dagger_{sql}\cdot b^\dagger_{sql}=0$$

Problem 2.4.7: The Kondo Problem:

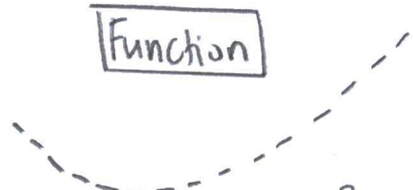


Cite: W. J. de Haas

"The electrical resistance of gold, copper, and lead at low temperatures"

Physical 1. 1115 (1934)

Function



$$\rho(T)=\rho_0+aT^2+C_m \ln\frac{u}{T}+bT^5$$

Cite: J. Kondo

"Resistance in dilute Magnetic alloys"

Prog. Theor. Phys. 32, 37 (1964)

Energy Function

$$E_{TOTAL}=E_{mag}+E_{repulsion}+E_{\substack{impurity\\ ratio}}$$

$$\hat{H}=H_{\circ f}+H_{\circ d}+(H_{corr}+H_{sd})$$

Cite: P. W. Anderson

"Localized magnetic States in metals"

Phys Rev. 124, 41 (1961)

(from problem) "Anderson impurity Hamiltonian"

$$\hat{H} = \sum_{k\sigma} (\epsilon_k c^{\dagger}_{k\sigma} c_{k\sigma} + (V_k d^{\dagger}_{\sigma} c_{k\sigma} + h.c.)) + \sum_{\sigma} \epsilon_d n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow}$$

where $c^{\dagger}_{k\sigma}$ = electron operator

σ = spin

ϵ_k = metallic host

$d^{\dagger}_{\sigma}$ = operator on impurity.

Again, the problem relates minimum energy to spin, orbitals, and Coulomb repulsion from boundary conditions in an amalgam.

$V_k = e^{ikd} \langle \phi_d | \hat{H} | \psi_d \rangle$ = coupling between states.

ϕ_d = atomic d-level

ψ_d = conducting d-level

$n_{d\sigma} = d^{\dagger}_{\sigma} d_{\sigma}$ = number operator

U = Coulomb interaction strength

$\epsilon_d < \epsilon_F < \epsilon_d + U$ - Fermi level

Minimum Hamiltonian:

Schrodinger equation: $\sum_{n=0}^{2} \hat{H}_{mn} |\psi_n\rangle = E|\psi_n\rangle$

Hamiltonian projection: $\hat{H}_{mn} = \hat{P}_m \hat{H} \hat{P}_n$

where P_m = Projection onto m-electron

$\hat{P}_0 = \Pi_\sigma (1 - n_{d\sigma})$

$\hat{P}_1 = (1 + n_{d\uparrow} + n_{d\downarrow} - 2n_{d\uparrow} n_{d\downarrow}$

$\hat{P}_2 = n_{d\uparrow} n_{d\downarrow}$

a) $\hat{H}_{mn} = \hat{P}_m \hat{H} \hat{P}_n$

$$= \hat{P}_m \begin{pmatrix} \epsilon_k c^\dagger_{k\sigma} c_{k\sigma} & V_d d^\dagger_\sigma (1-n_d) c_{k\sigma} & 0 \\ V_d d_\sigma (1-n^\dagger_{d\sigma}) c^\dagger_{k\sigma} & \epsilon_k c^\dagger_{k\sigma} c_{k\sigma} + \epsilon_d & V_k d^\dagger_\sigma n_{d\sigma} c_{k\sigma} \\ 0 & V_k d_\sigma n^\dagger_{d\sigma} c^\dagger_{k\sigma} & \epsilon_k c^\dagger_{k\sigma} c_{k\sigma} + 2\epsilon_d + U \end{pmatrix} \hat{P}_n$$

Coupling involves $n=1, m=1$

electrons in

$$\hat{H}_{mn} = \hat{P}_m \begin{pmatrix} H_{00} & H_{01} & H_{02} \\ H^\dagger_{10} & H_{11} & H_{12} \\ H^\dagger_{20} & H^\dagger_{21} & H_{22} \end{pmatrix} \hat{P}_n$$

So, $\hat{H}_{20} = \hat{H}_{02} = 0$

While $\hat{H}_{12} = V_k d^\dagger_\sigma n_{d\sigma} c_{k\sigma}$

$\hat{H}_{10} = V_d d^\dagger_\sigma (1 - n_d) c_{k\sigma}$

b) Schrodinger Equation:

$$H_{mn}|\psi\rangle = E|\psi\rangle$$

$$\begin{pmatrix} \hat{H}_{00} & \hat{H}_{01} & 0 \\ \hat{H}_{10} & \hat{H}_{11} & \hat{H}_{12} \\ 0 & \hat{H}_{21} & \hat{H}_{22} \end{pmatrix} \begin{pmatrix} |\psi_0\rangle \\ |\psi_1\rangle \\ |\psi_2\rangle \end{pmatrix} = E \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \\ |\psi_3\rangle \end{pmatrix}$$

$$\hat{H}_{00}|\psi_0\rangle + \hat{H}_{00}|\psi_1\rangle = E|\psi_0\rangle$$

$$|\psi_0\rangle = \frac{\hat{H}_{01}|\psi_1\rangle}{E - H_{01}}$$

$$\hat{H}_{21}|\psi_1\rangle + H_{22}|\psi_2\rangle = E|\psi_2\rangle$$

$$|\psi_2\rangle = \frac{\hat{H}_{21}|\psi_1\rangle}{E - H_{22}}$$

$$H_{10}|\psi_0\rangle + H_{11}|\psi_1\rangle + H_{12}|\psi_2\rangle = E|\psi_1\rangle$$

$$\left[\hat{H}_{10}\frac{1}{E - H_{00}}H_{01} + H_{11} + H_{12}\frac{1}{E - H_{22}}H_{21}\right]|\psi_1\rangle = E|\psi_1\rangle$$

b) $$\hat{H}_{12} = \hat{H}_{12}\frac{1}{E - H_{22}}\hat{H}_{21}$$

$$= \sum_{k k' \sigma \sigma'} V_k V_{k'}^{*} \, c_{k\sigma} n_{d\sigma} d_\sigma \frac{1}{E - \epsilon_k + 2\epsilon_d - V} d_\sigma^{\dagger} n_{d\sigma} c_{k\sigma}$$

$$\hat{H}_{10} = \hat{H}_{10}\frac{1}{E - H_{00}}\hat{H}_{01}$$

$$= \sum_{kk'\sigma\sigma'} V_k V_k d_\sigma^\dagger (1-n_{d\sigma}) C_{k\sigma} \frac{1}{E - \epsilon_k} C_{k\sigma}^\dagger (1-n_{d\sigma'}) d_{\sigma'}$$

$$\hat{H} = \sum_{kk'\sigma\sigma'} V_k V_k \left(\frac{C_{k'\sigma} C_{k\sigma}^\dagger d_\sigma d_\sigma^\dagger}{U + \epsilon_d - \epsilon_{k'}} + \frac{C_{k'\sigma} C_{k\sigma}^\dagger d_{\sigma'} d_\sigma}{\epsilon_k - \epsilon_d} \right)$$

∴ when $\epsilon_k \approx E$

$$\simeq \sum_{kk'\sigma\sigma'} V_k V_{k'} \left(2\, \hat{S}_{kk} \cdot \hat{S}_d + \frac{1}{2} \sum C_{k\sigma}^\dagger C_{k'\sigma} n_{d\sigma} \right)$$

$$\times \left(\frac{1}{U + \epsilon_d - \epsilon_{k'}} - \frac{1}{\epsilon_k - \epsilon_d} \right)$$

$$\simeq \sum_{kk'} \left[2 \cdot J_{kk} \cdot \hat{S}_{kk} \cdot \hat{S}_d + K_{kk'} \sum C_{k\sigma}^\dagger C_{k'\sigma} \right]$$

Where $J_{kk'}$ = Exchange interaction

$K_{kk'}$ = Scattering term

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k C_{k\sigma}^\dagger C_{k\sigma} + \hat{H}$$

$$= \sum_{k\sigma} \epsilon_k C_{k\sigma}^\dagger C_{k\sigma} + 2 \cdot J\, \hat{S}_{kk} \hat{S}_d$$

∴ when scattering ($K_{kk'} = 0$)